

# Yuchong Pan

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| CONTACT<br>INFORMATION | MIT, Room 2-341A<br>77 Massachusetts Avenue<br>Cambridge, MA 02139<br>yuchong@mit.edu<br>ORCID 0000-0002-6134-4232                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| RESEARCH<br>INTERESTS  | Algorithms, combinatorics, optimization, theoretical computer science, graph theory, operations research.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| EDUCATION              | <b>Massachusetts Institute of Technology</b><br>Ph.D., Applied Mathematics, expected 2027 <ul style="list-style-type: none"><li>◦ Advisor: Michel X. Goemans</li></ul> <b>University of British Columbia</b><br>B.Sc., Computer Science and Mathematics, Honours, with Distinction, 2021 <ul style="list-style-type: none"><li>◦ Thesis: <i>Optimization Problems on Network Flows with Degree Constraints</i></li><li>◦ Advisor: F. Bruce Shepherd</li></ul>                                                                                                                                                                                                                                                                                                                                                                                         |
| EMPLOYMENT             | <b>Microsoft Corporation</b><br>Software Engineer Intern, May 2020–Aug 2020<br>Software Engineer Intern, Jun 2019–Aug 2019<br>Software Engineer Intern, May 2018–Aug 2018<br><b>Sogou, Inc.</b><br>Software Engineer Intern, May 2017–Jul 2017<br><b>InitialView</b><br>Software Engineer Intern, Aug 2016–May 2017                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| PUBLICATIONS           | <ul style="list-style-type: none"><li>◦ Bicriteria approximation algorithms for demand matching (with M. X. Goemans). <i>The 2027 SIAM Symposium on Simplicity in Algorithms (SOSA 2027)</i>, to appear. <i>arXiv preprint arXiv:2608.04223</i>, <a href="https://arxiv.org/abs/2608.04223">https://arxiv.org/abs/2608.04223</a>.</li><li>◦ Planarity via spanning tree number: a linear-algebraic criterion (with A. Bu). <i>SIAM J. Discrete Math.</i>, 39 (2025), pp. 728–751, <a href="https://doi.org/10.1137/24M1660395">https://doi.org/10.1137/24M1660395</a>.</li></ul>                                                                                                                                                                                                                                                                      |
| MANUSCRIPTS            | <ul style="list-style-type: none"><li>◦ A better-than-3 approximation algorithm for demand matching via knapsack intersection LP and contention resolution (with M. X. Goemans). <i>arXiv preprint arXiv:2609.17932</i>, <a href="https://arxiv.org/abs/2609.17932">https://arxiv.org/abs/2609.17932</a>.</li><li>◦ On high-value and high-flow cycles at basic feasible solutions of subtour elimination relaxations for the symmetric and asymmetric traveling salesman problems (with M. X. Goemans). <i>In preparation</i>.</li><li>◦ A counterexample to box-half-integrality of the intersection of crossing submodular flow systems (with M. X. Goemans). 2023. <i>In preparation</i>.</li><li>◦ Optimization problems on network flows with degree constraints. Undergraduate honours thesis, University of British Columbia, 2021.</li></ul> |

## TALKS AND PRESENTATIONS

- A better-than-3 approximation algorithm for demand matching. *Theory Lunch*. Computer Science and Artificial Intelligence Laboratory (CSAIL), Massachusetts Institute of Technology. Cambridge, MA. September 29, 2026.
- Bicriteria approximation algorithms for demand matching. *Simple Person's Applied Math Seminar (SPAMS)*. Department of Mathematics, Massachusetts Institute of Technology. Cambridge, MA. September 10, 2026.
- Bicriteria approximation algorithms for demand matching. *The 9th YinzOR Student Conference*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. August 21, 2026.
- Planarity via spanning tree number: a linear-algebraic criterion. *The 2025 AMS New England Graduate Student Conference*. Brown University. Providence, RI. April 19, 2025.
- The tales of planar graphs: the extremal problems, the criterion, and the conjectures. *Meet-and-Think*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. February 19, 2025.
- Fantastic planar graphs and where to find them: a linear-algebraic planarity criterion. *Theory Lunch*. Computer Science and Artificial Intelligence Laboratory (CSAIL), Massachusetts Institute of Technology. Cambridge, MA. February 10, 2025.
- From submodular flows to planar graphs: a planarity criterion inspired by an OR question. *The 7th YinzOR Student Conference*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. August 23, 2024.
- Planarity via spanning tree number: a linear-algebraic criterion. *Algorithms Seminar*. Department of Computer Science, University of British Columbia. Vancouver, BC. May 23, 2024.
- Fantastic spanning trees in planar graphs and where to find them. *Simple Person's Applied Math Seminar (SPAMS)*. Department of Mathematics, Massachusetts Institute of Technology. Cambridge, MA. October 12, 2023.
- Perturbation-stable maximum cuts. *Algorithms Reading Group*. Department of Computer Science, University of British Columbia. Online. June 30, 2020.

## TEACHING EXPERIENCE

### Massachusetts Institute of Technology

#### Teaching Assistant (Grader)

|             |                                                                                  |
|-------------|----------------------------------------------------------------------------------|
| 18.435J     | Quantum Computation, Fall 2026<br><i>Same subject as 2.111J, 6.6410J, 8.370J</i> |
| 18.453/4531 | Combinatorial Optimization, Spring 2023                                          |
| 18.085/0851 | Computational Science and Engineering I, Fall 2022                               |

#### Instructor-G (Course Administrator & Recitation Instructor)

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| 18.06 | Linear Algebra, Spring 2026 |
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#### Teaching Assistant (Research Mentor)

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| 18.821 | Project Laboratory in Mathematics, Fall 2025 |
|--------|----------------------------------------------|

#### Teaching Assistant (Recitation Instructor)

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| 18.600 | Probability and Random Variables, Spring 2025 |
| 18.06  | Linear Algebra, Spring 2024                   |

### University of British Columbia

#### Teaching Assistant

|              |                                                      |
|--------------|------------------------------------------------------|
| CPSC 420     | Advanced Algorithms Design and Analysis, Spring 2021 |
| CPSC 311     | Definition of Programming Languages, Fall 2020       |
| CPSC 421/501 | Introduction to Theory of Computing, Fall 2019       |
| CPSC 121     | Models of Computation, Fall 2018                     |

*Academic Assistant*

CPSC 411

Introduction to Compiler Construction, Fall 2019–Spring 2020

*Assisted course redesign, supervised by William J. Bowman.*

MENTORING  
EXPERIENCE

**Massachusetts Institute of Technology**

Directed Reading Program (DRP), Jan 2025–Feb 2025.

- Students: Tiago Oliveira Marques, Eric Shen, Jophy Ye
- Book: *Graphs and Geometry* by László Lovász

Program for Research in Mathematics, Engineering and Science for High School Students (PRIMES-USA, remote), Jan 2024–Dec 2024.

- Students: Christopher K. Bao (Davidson Academy, Reno, NV), Joshua T. Wang (Brookfield Central High School, Brookfield, WI), William Zhao (Dougherty Valley High School, San Ramon, CA)
- Topic: Approximate Minimum Cuts in Graphs and Hypergraphs
- A poster titled *A Faster Approximation Algorithm for the  $k$ -Way Cut Problem in Low-Rank Hypergraphs via Multiplicative Weight Update* was presented by Christopher K. Bao in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- A poster titled *Improved Analysis of the Branching Contraction Algorithm for the Unweighted Hypergraph Minimum  $k$ -Way Cut Problem* was presented by Joshua T. Wang in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- A poster titled *Upper and Lower Bounds on Several Cut Numbers in Hypergraphs via Randomized Contraction* was presented by William Zhao in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- Christopher K. Bao, Joshua T. Wang, and William Zhao attend Massachusetts Institute of Technology after high school.

Directed Reading Program (DRP), Jan 2024–Feb 2024.

- Students: Jennifer Ai, Neha Pant
- Book: *Algorithmic Game Theory* edited by N. Nisan, T. Roughgarden, É. Tardos and V. V. Vazirani

Program for Research in Mathematics, Engineering and Science for High School Students (PRIMES-USA, remote), Jan 2023–Dec 2023.

- Student: Ruiyang (Raymond) Luo (University High School, Irvine, CA)
- Topic: Cyclic Basis Orderings and Equitability of Matroids
- Ruiyang (Raymond) Luo attends Massachusetts Institute of Technology after high school.

Research Science Institute (RSI), Jun 2023–Aug 2023.

- Student: Alan Bu (Phillips Exeter Academy, Exeter, NH)
- Topic: Connections Between Concatenations of Two Incidence Matrices and Spanning Trees in Planar Graphs
- This project was selected as one of the top 5 out of 100 oral presentations.
- Alan Bu won the 10th place (with a \$40,000 scholarship) in the 83rd Regeneron Science Talent Search (STS) for this project.
- Alan Bu attends Harvard University after high school.

Research Science Institute (RSI), Jun 2023–Aug 2023.

- Student: Deyan Hadzhi-Manich (High School of Mathematics “Dr. Petar Beron”, Varna, Bulgaria)
- Topic: Lower and Upper Bounds on Longest Geometrically Increasing Sequences
- Deyan Hadzhi-Manich attends Harvard University after high school.

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|                       | <p>Directed Reading Program (DRP), Jan 2023–Feb 2023.</p> <ul style="list-style-type: none"> <li>◦ Students: Tianze Jiang, Wayne Zhao</li> <li>◦ Book: <i>Beyond the Worst-Case Analysis of Algorithms</i> edited by T. Roughgarden</li> </ul> <p>Extended Summer Program in Undergraduate Research (SPUR+), Jun 2022–Aug 2022.</p> <ul style="list-style-type: none"> <li>◦ Students: Evgeniya Artemova, Christina Yu</li> <li>◦ Topic: Lower Bounds on the Discrete Newton’s Algorithm for the Submodular Line Search Problem</li> </ul> <p>Directed Reading Program (DRP), Jan 2022–Feb 2022.</p> <ul style="list-style-type: none"> <li>◦ Students: Sualeh Asif, Aparna Gupte</li> <li>◦ Book: <i>Randomized Algorithms</i> by P. Raghavan and R. Motwani</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| HONORS AND AWARDS     | <ul style="list-style-type: none"> <li>◦ Graduation with Distinction, University of British Columbia, 2021.</li> <li>◦ Science Scholar and Dean’s Honour List, University of British Columbia, 2018–2021.</li> <li>◦ Scholarships at UBC (total CAD\$51,550): Work Learn International Undergraduate Research Award, 2021; Stanley M Grant Scholarship in Mathematics, 2019, 2021; Trek Excellence Scholarship, 2018–2020; Faculty of Science International Student Scholarship, 2018–2020; J Fred Muir Memorial Scholarship in Science, 2020; Dean of Science Scholarship, 2018–2019; Marie Kendall Memorial Scholarship in Science, 2018; Joel Harold Marcoe Memorial Scholarship, 2018; Outstanding International Student Award, 2017.</li> <li>◦ 27th Place (out of 118 teams), North American Invitational Programming Contest, Open Division (USA + Canada), 2018.</li> <li>◦ 11th Place (out of 67 teams), ACM International Collegiate Programming Contest, Pacific Northwest Regional (Division 1), 2017.</li> <li>◦ Silver Medal, IOI China Team Selection Competition, China Computer Federation, 2015.</li> <li>◦ Bronze Medal, Asia-Pacific Informatics Olympiad, China Computer Federation, 2015.</li> <li>◦ First Prize, National Olympiad in Informatics in Provinces (Grades 10–12 Division), China Computer Federation, 2013–2014.</li> <li>◦ Second Prize, National Olympiad in Informatics in Provinces (Grades 7–9 Division), China Computer Federation, 2011–2012.</li> </ul> |
| PROFESSIONAL SERVICES | <p>Journal Review: SIAM J. Discrete Math., Math. Program., Oper. Res. Lett.</p> <p>Conference Review: IPCO 2024, SODA 2022</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| RELEVANT SKILLS       | <p>Languages: English, Mandarin</p> <p>Programming: <math>\text{\LaTeX}</math>, Python, Julia, C/C++, Java, C#, MATLAB, JavaScript, Go, Racket, Standard ML, MySQL</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| LAST UPDATED          | October 1, 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |